

2.4.1 Continuity

Motivation: We want to call a function *continuous* if we can draw the graph of the function without picking up our writing utensil.

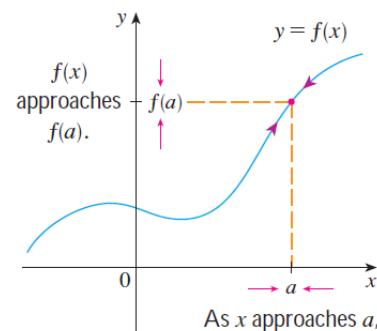
In class we will sketch some examples of continuous/discontinuous functions.

We now present a rigorous definition of continuity of a function at a point.

Definition: A function f is **continuous at a number a** if $\lim_{x \rightarrow a} f(x) = f(a)$.

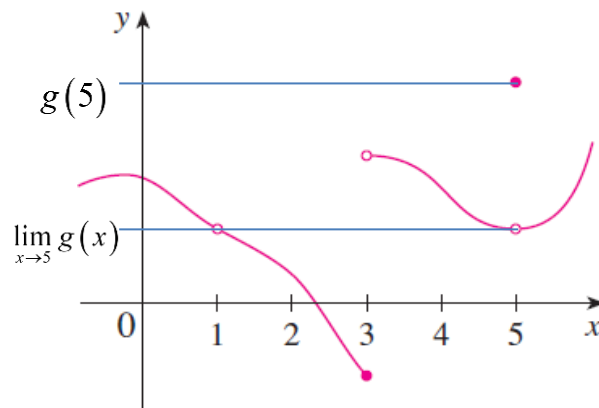
Note: It is very important to note that if one or both sides of the equation defining continuity does not exist, then the function is NOT continuous at a ! For example, if the limit is infinite, the function is not continuous at a .

The following picture illustrates the definition of continuity above:



Example 1: The function $g(x)$ is pictured below. Would you say that g is continuous at 5?

No, based upon the picture it seems that $\lim_{x \rightarrow 5} g(x) \neq g(5)$.



Question: What are all of the values of x at which the function $g(x)$ is discontinuous?

$g(x)$ is discontinuous at $x = 1, 3, \& 5$

Example 2: State whether the following function is continuous at 1. Explain your answer.

$$f(x) = \begin{cases} \frac{x-1}{x^2-1} & \text{if } x \neq 1 \\ \frac{1}{2} & \text{if } x = 1 \end{cases}$$

We note that $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{x-1}{x^2-1} = \lim_{x \rightarrow 1} \frac{(x-1)}{(x-1)(x+1)} = \lim_{x \rightarrow 1} \frac{1}{x+1} = \frac{1}{2}$.

Also, $f(1) = \frac{1}{2}$.

Thus, $\lim_{x \rightarrow 1} f(x) = f(1)$, and the function is continuous at 1 by definition.

Example 3: Prove that the function $f(x) = \begin{cases} \frac{x^2}{4-x} & \text{if } x \geq 5 \\ 10-7x & \text{if } x < 5 \end{cases}$ is continuous at 5.

We need to show that $\lim_{x \rightarrow 5} f(x) = f(5)$. In order for this to be true we need to have the limit exist. Since we can approach 5 using each of the pieces of f we must check the limit from both sides.

We note that $\lim_{x \rightarrow 5^-} f(x) = \lim_{x \rightarrow 5^-} (10-7x) = 10-7(5) = -25$ and that $\lim_{x \rightarrow 5^+} f(x) = \lim_{x \rightarrow 5^+} \frac{x^2}{4-x} = \frac{(5)^2}{4-5} = -25$.

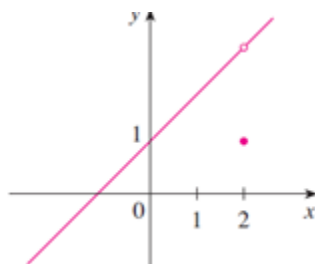
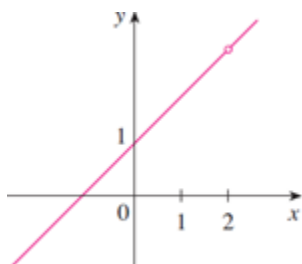
Therefore we can say that $\lim_{x \rightarrow 5} f(x) = -25$

Also, $f(5) = \frac{(5)^2}{4-5} = -25$.

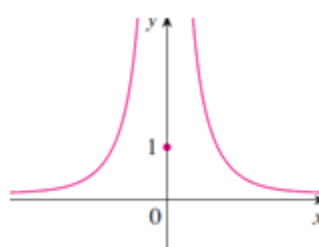
Thus, $\lim_{x \rightarrow 5} f(x) = f(5)$, and the function is continuous at 5 by definition.

Types of Discontinuities

Removable Discontinuities



Infinite Discontinuities

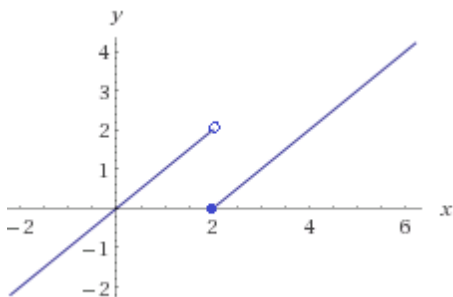


Jump Discontinuities



Example 4: Find all the real numbers at which the following function is discontinuous. Also state the type of discontinuity at each number that you find: $h(x) = \begin{cases} x-2 & \text{if } x \geq 2 \\ x & \text{if } x < 2 \end{cases}$.

We begin by graphing the function:



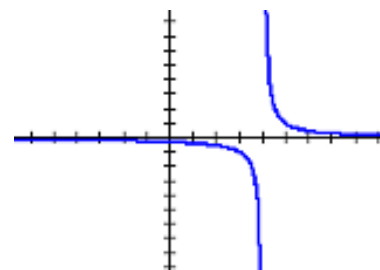
We see (intuitively) from the graph that there is a discontinuity only at 2. Specifically, $\lim_{x \rightarrow 2} h(x)$ DNE. Moreover, the discontinuity at 2 is a jump discontinuity.

Example 5: Answer each of the following.

a) Give an example of a function with an infinite discontinuity at 4.

There are many possible examples. We just need at least one of the one sided limits as we approach 4 to be positive or negative infinity.

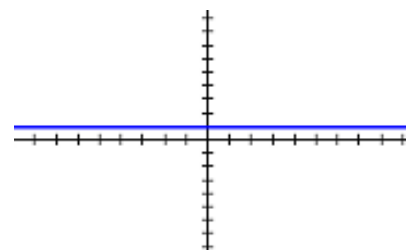
One example is $f(x) = \frac{1}{x-4}$ since $\lim_{x \rightarrow 4^+} f(x) = \infty$.



b) Give an example of a function with a removable discontinuity at 4.

For a removable discontinuity we need the limit of the function as we approach 4 to exist but not equal the function evaluated at 4.

One example is $g(x) = \frac{x-4}{x-4}$. Note that $\lim_{x \rightarrow 4} g(x) = 1$, yet $g(4)$ is undefined.



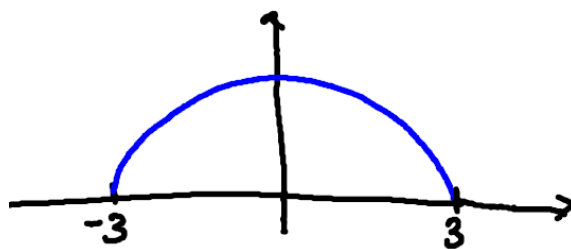
We now introduce some definitions to help us make sense of statements like: “ f is continuous on an open/closed interval”.

Definition: A function f is continuous from the right at a if $\lim_{x \rightarrow a^+} f(x) = f(a)$, and f is continuous from the left at a if $\lim_{x \rightarrow a^-} f(x) = f(a)$.

Definitions:

1. A function f is continuous on the open interval (a,b) if f is continuous at each number in the interval.
2. A function f is continuous on the closed interval $[a,b]$ if the following three conditions hold:
 - (a) f is continuous on (a,b)
 - (b) f is continuous from the right at a .
 - (c) f is continuous from the left at b .

Example 6: Consider the function $f(x) = \sqrt{9-x^2}$. The graph of the function is below:



- a) State the domain of f .

We need $9-x^2 \geq 0 \Rightarrow -3 \leq x \leq 3$. So the domain is all the values of x in $[-3, 3]$. This is also clear from the graph of the function.

- b) Show that $f(x) = \sqrt{9-x^2}$ is continuous on its domain.

We must verify the three conditions of the most recent definition. First note that for any c in $(-3, 3)$, we can obtain the following from the limit laws: $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} \sqrt{9-x^2} = \sqrt{9-c^2} = f(c)$. This shows that f is continuous on $(-3, 3)$.

Again using the limit laws, we note that:

$$\lim_{x \rightarrow -3^+} f(x) = \lim_{x \rightarrow -3^+} \sqrt{9-x^2} = \lim_{x \rightarrow -3^+} \sqrt{9-(-3)^2} = 0 = f(-3), \text{ and}$$

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} \sqrt{9-x^2} = \lim_{x \rightarrow 3^-} \sqrt{9-3^2} = 0 = f(3).$$

So, f is right continuous at -3 and left continuous at 3 .

We conclude that f is continuous on the closed interval $[-3, 3]$, and hence f is continuous on its domain. ■

We now present two theorems which will help us determine when functions that we are familiar with from Precalculus are continuous.

Theorem: If f and g are continuous at a and c is constant, then the following functions are also continuous at a :

1. $f + g$
2. $f - g$
3. cf
4. fg
5. $\frac{f}{g}$ if $g(a) \neq 0$

Proof (of 1):

We know that since f and g are continuous at a , $\lim_{x \rightarrow a} f(x) = f(a)$ and $\lim_{x \rightarrow a} g(x) = g(a)$. So, using the first limit law, we have that: $\lim_{x \rightarrow a} (f + g)(x) = \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = f(a) + g(a) = (f + g)(a)$. ■

Note: The proof of statements 2, 3, 4, and 5 are similar to the proof of statement 1.

Theorem: The following types of functions are continuous at every number in their domains.

- Polynomial Functions
- Rational Functions
- Root Functions
- Trigonometric Functions
- Inverse Trigonometric Functions
- Exponential Functions
- Logarithmic Functions

Example 7: Use the two most recent theorems to determine where the function: $f(x) = \frac{\ln(x) + \arctan(x)}{x^2 - 1}$ is continuous.

We first note that domain of $r(t) = \ln(t)$ is all positive real numbers, and the domain of $\arctan(x)$ is all real numbers. Thus, by the two most recent theorems $h(x) = \ln(x) + \arctan(x)$ is continuous on the interval $(0, \infty)$. Moreover, the function $g(x) = x^2 - 1$ is continuous on the set of real numbers, and $g(\pm 1) = 0$.

So, it follows that the function $f(x)$ is continuous on $(0, 1) \cup (1, \infty)$. In words, $f(x)$ is continuous on the set of positive real numbers with one excluded.

Note: Another way of writing the inverse tangent function is $\tan^{-1}(x)$. One should note that this is NOT to be interpreted as tangent to the negative first power. We write tangent to the negative first power as $(\tan(x))^{-1}$.

Example 8: Find a value of C that makes g continuous on $(-\infty, \infty)$.

$$g(x) = \begin{cases} x+1, & x \leq 1 \\ e^{1-x} + C, & x > 1 \end{cases}$$

We know that g is continuous on the interval $(-\infty, 1)$ because $x+1$ is a polynomial function.

We know that g is continuous on the interval $(1, \infty)$ because $e^{1-x} + C$ is the sum of an exponential function and a polynomial function.

We must find C so that g is continuous at $x = 1$. In particular, we need $\lim_{x \rightarrow 1} g(x) = g(1)$.

We note that $g(1) = 1+1 = 2$ and $\lim_{x \rightarrow 1^-} g(x) = \lim_{x \rightarrow 1^-} (x+1) = 1+1 = 2$. Moreover,

$$\lim_{x \rightarrow 1^+} g(x) = \lim_{x \rightarrow 1^+} (e^{1-x} + C) = e^{1-1} + C = 1 + C$$

If $C = 1$, then $\lim_{x \rightarrow 1^+} g(x) = 2$ and therefore we would have $\lim_{x \rightarrow 1} g(x) = 2$

Since $\lim_{x \rightarrow 1} g(x) = g(1)$ we can say that g is continuous at $x = 1$. Therefore g is continuous on $(-\infty, \infty)$.

Example 9: Consider the function: $f(x) = \frac{\sin(x)}{\cos(x)+2}$.

a) State where this function is continuous.

We note that the domain of the sine function is the set of real numbers. So, the sine function is continuous everywhere. Similarly, $g(x) = \cos(x) + 2$ is the sum of two functions continuous everywhere. So, g is continuous on all real numbers. Moreover, since $\cos(x) > -2$ for all values of x , g is never 0. Therefore, the function in question is continuous on the set of real numbers.

b) Evaluate $\lim_{x \rightarrow \frac{\pi}{3}} \frac{\sin(x)}{\cos(x)+2}$ (be exact).

By part (a) the function is continuous at $\frac{\pi}{3}$. So, $\lim_{x \rightarrow \frac{\pi}{3}} \frac{\sin(x)}{\cos(x)+2} = f\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}/2}{1/2+2} = \frac{\sqrt{3}}{5}$.

Note: We are more or less using a generalization of the direct substitution property from section 2.3.1 in (b). In particular, if we are trying to find $\lim_{x \rightarrow a} f(x)$, and we know that f is continuous at a , then the limit can be found by simply plugging a in for x (i.e. finding $f(a)$).

Question to the Class: How might you approach evaluating the limit: $\lim_{x \rightarrow 1} \arcsin\left(\frac{1-\sqrt{x}}{1-x}\right)$?

Answer (partial): One might hope to bring the limit inside the inverse sine function, but can we do this? In particular, is it true that $\lim_{x \rightarrow 1} \arcsin\left(\frac{1-\sqrt{x}}{1-x}\right) = \arcsin\left(\lim_{x \rightarrow 1} \frac{1-\sqrt{x}}{1-x}\right)$? The next theorem addresses this.

Theorem: If f is continuous at b and $\lim_{x \rightarrow a} g(x) = b$, then $\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right) = f(b)$.

Note: The theorem above immediately implies that if g is continuous at a and f is continuous at $g(a)$, then $(f \circ g)(x) = f(g(x))$ is continuous at a as well.

Example 10: We can use the theorem above to evaluate the limit mentioned in the question above. Specifically,

$$\begin{aligned} \lim_{x \rightarrow 1} \arcsin\left(\frac{1-\sqrt{x}}{1-x}\right) &= \arcsin\left(\lim_{x \rightarrow 1} \frac{1-\sqrt{x}}{1-x}\right) = \arcsin\left(\lim_{x \rightarrow 1} \frac{1-\sqrt{x}}{1-(\sqrt{x})^2}\right) \\ &= \arcsin\left(\lim_{x \rightarrow 1} \frac{1-\sqrt{x}}{(1-\sqrt{x})(1+\sqrt{x})}\right) = \arcsin\left(\lim_{x \rightarrow 1} \frac{1}{1+\sqrt{x}}\right) = \arcsin\left(\frac{1}{2}\right) = \frac{\pi}{6}. \end{aligned}$$

Example 11 (if time permits): Where is the function $f(x) = \ln(1 - \cos(x))$ continuous?

We know that the natural log function is continuous on the set of positive real numbers, and $h(x) = 1 - \cos(x)$ is continuous on the set of real numbers. So, the function will be continuous as long as $1 - \cos(x) > 0$. We note that $1 - \cos(x) \geq 0$ and $1 - \cos(x) = 0$ if and only if $x = 0, \pm 2\pi, \pm 4\pi, \dots$. Thus, the function is discontinuous when x is an even multiple of pi, and it is continuous on the intervals between these values.

Note: This is particularly impressive, because this function has infinitely many discontinuities. One can confirm this solution with the graphing calculator.